Teaching Logion How Scribes Actually Slip:
A Weighted Edit Distance for Byzantine Greek Emendation

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Extends Logion (Cowen-Breen, Brooks, Graziosi, Haubold; *ACL ALP-2023*)

Abstract

Princeton's Logion is a BERT-based model that flags likely scribal errors in premodern Greek and proposes corrections. To decide which spellings count as plausible alternatives, it uses classical Levenshtein distance, which treats every single-character substitution as equally costly. That assumption is wrong for Byzantine Greek, where a small set of phonetic and visual confusions causes most copying errors. Replacing the binary distance-1 filter with a *weighted* Levenshtein neighborhood — substitution costs drawn from documented confusion classes — gives a statistically significant gain on real attested manuscript variants: top-5 recovery of 42.3% versus a baseline of 35.7% on 333 loci from the SBLGNT apparatus (paired McNemar $p < 10^{-3}$). Two other tested components are inert or harmful, so the winning configuration is the simple one: weighted Levenshtein on, everything else off.

1 What Logion does

Ancient Greek survives through centuries of hand-copying, and the copies are full of errors: smudged letters, dropped words, sounds misheard from dictation. Logion is a language model (a BERT fine-tuned on premodern Greek) that reads a passage, flags positions where the transmitted word looks wrong, and proposes the word it thinks belongs there. Editors use it as a second reader.

For each word it computes a *chance-correction ratio*: how probable is the word actually written, compared with the most probable near-spelling the model can substitute in context? Words whose written form is far less probable than an available alternative are surfaced for expert review. Building that set of "near-spellings" requires a notion of which words count as close, and for this Logion uses **edit distance** (Levenshtein): the number of single-character changes that turn one word into another.

2 The problem: every substitution costs the same

Classical edit distance is blunt. Changing one letter to *any* other letter costs exactly one, whether the two are routinely confused or wholly unrelated. Logion's original filter is blunter still: a yes/no gate that admits any vocabulary word within distance 1 and weights them all equally.

Real copying errors are not uniform. In Byzantine Greek a handful of confusions dominate:

- **Itacism** — η, ι, υ, ει, οι all came to be pronounced /i/, so scribes writing from dictation swapped them constantly. This is the single largest source of error.
- **Visual look-alikes** in minuscule script — γ/τ, λ/μ, β/ν, ρ/π are easy to misread.
- **Post-vocalic** β/υ — a sound shift that bled into spelling.

A metric that treats η→ι the same as η→ξ discards exactly the signal that matters.

3 The change: weight the substitutions

Instead of a yes/no gate, give each substitution a *cost* taken from documented confusion classes (Allen, *Vox Graeca*; West, *Textual Criticism*; Reynolds & Wilson, *Scribes & Scholars*). Easily-confused letters cost little; unrelated letters cost the full amount. The filter becomes a continuous weight rather than a hard cutoff. An itacism swap such as ταις / τοις costs 0.35 and receives about 3.7× the weight of an unrelated same-distance edit such as τις / οις, which costs the full 1.00.

4 Result

The fairest test uses **real attested variants**: 333 loci in the SBLGNT apparatus where the manuscripts genuinely disagree, i.e. errors real scribes actually made rather than artificial ones. There the weighted filter recovers more of them.

> **Top-5 recovery:** 42.3% **vs.** 35.7% **baseline** (paired McNemar $p < 10^{-3}$). The weighted filter recovered 28 loci the baseline missed and lost only 6. Top-1 trends positive but is not significant (16.5% vs. 14.4%, $p = 0.30$).

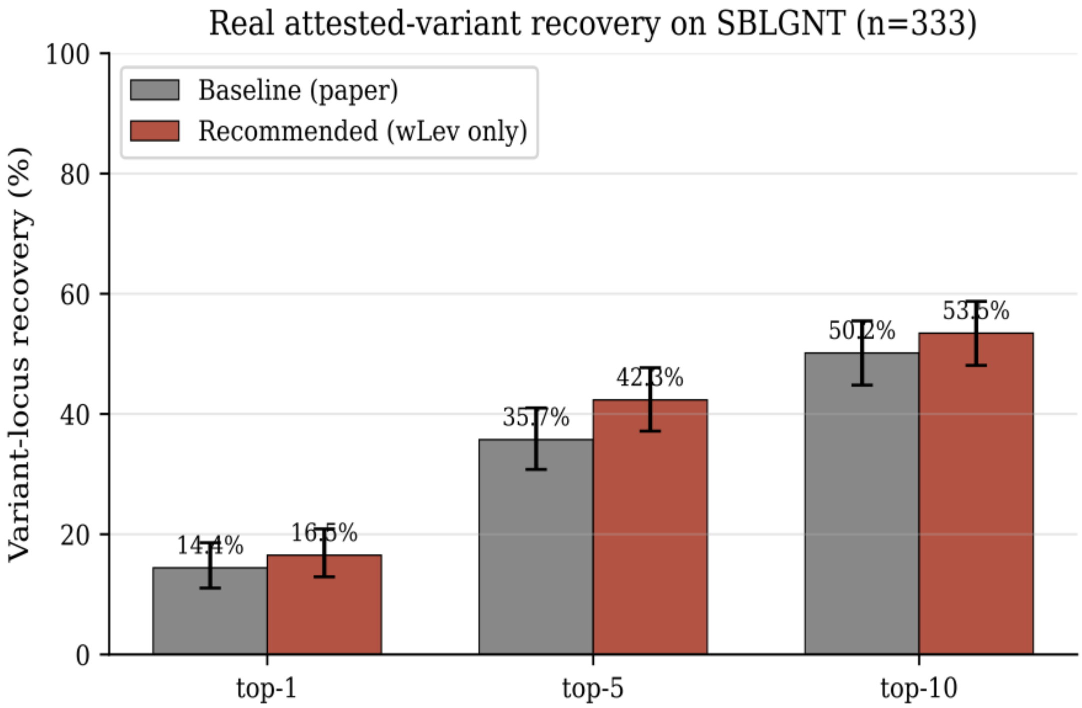

Figure 1: Real attested-variant recovery on SBLGNT ($n = 333$). Recommended = weighted Levenshtein only. Error bars are Wilson 95% confidence intervals; the top-5 gap is statistically significant.

The same direction appears in two more controlled tests. On the paper's synthetic protocol ($n = 200$), removing the weighted filter costs -4.0pp at top-1 (one-tailed McNemar $p = 0.029$). On a corruption protocol built to mimic scribal-class errors, removing it costs -4.0pp at top-10 ($p = 0.039$).

5 What helped, what didn't

Two further components were tested alongside the weighted distance. Both proved to be dead weight, and the all-on combination does *worse* than baseline (-1.5pp top-1, -3.5pp top-5).

- A **chance-fit adaptive temperature** (flatten the model's distribution when the written word is implausible) is inert — an exact tie with baseline at top-1, McNemar $p = 1.00$.
- Four **extras (E1–E4)** are actively harmful on the realistic protocol: removing them lifts top-1 from 26.5% to 30.5% ($p = 0.021$). The likely culprit is the *lectio difficilior* penalty (E1), which discounts suggestions more common than the written word — exactly the pattern when an itacism slip moves a rare spelling to a common one.

The winning configuration is therefore the simple one: **weighted Levenshtein on, everything else off.**

Table 1: Per-component ablation on artificial single-character corruption ($n = 200$, §4.1 protocol).

Condition	Top-1	Top-5	Top-10	McNemar p
Baseline (paper)	37.5%	69.5%	81.5%	—
−CF temperature	37.5%	69.0%	80.0%	1.00
−**Weighted Lev**	**33.5**%	67.5%	80.5%	**0.057**
−Extras (E1–E4)	37.0%	68.0%	81.5%	1.00
All improvements	36.0%	66.0%	79.0%	0.63

6 Out-of-corpus deployment: Photius

The system was then run on 26 paragraphs (~2,100 words) of Photius's *Bibliotheca* (codices 70, 92, 161, 165, 213), a text outside Logion's training corpus and surviving in two manuscript families (Marcianus gr. 450 and 451). Its shortlist clusters in the itacism and visual-confusion region, as expected, but none of its picks align with documented apparatus loci. They are best read as positions where the model and the transmitted text most diverge, not as a curated emendation set.

7 Caveats

- **Domain shift.** The public checkpoint is fine-tuned on Psellus (11th c. Byzantine). The variant test uses Koine (1st c.) and the Photius run is 9th c., so absolute accuracy sits below the paper's in-domain 90.5%. The relative deltas — baseline vs. weighted on the same loci — still hold.
- **The synthetic protocol is rigged against weighting.** It draws corruptions uniformly across the 24-letter alphabet, which quietly rewards an algorithm that treats every edit as equally likely. Weighting exists to exploit the non-uniform reality of scribal error, so the real-variant test is the one that counts.
- **A variant is not always an error.** Many SBLGNT variants are disagreements between defensible readings, not slips. The test measures whether Logion finds a locus unusual, which is weaker than asking whether it is wrong.
- **The weights are hand-tuned** from textbooks. The natural next step is to learn the phonetic prior from Byzantine palaeographic data, or to retrain the model with the weighted neighborhood as a training signal rather than only an inference-time filter — which could internalize scribal error patterns directly and yield a larger gain.

Built on Logion (Cowen-Breen et al., *ACL ALP-2023*). SBLGNT apparatus from `jjmccollum/sblgnt-tei` (CC BY 4.0). Source: `github.com/Zed-Rez/logion-wlev`.